HERIOT WATT UNIVERSITY

SCHOOL OF MATHEMATICAL AND COMPUTER SCIENCES

Actuarial Mathematics and Statistics

F78PB

Probability and Statistics B

IN-PERSON EXAMINATION

April Examination and Assessment Diet 2023

Duration: 2 hours

SOLUTIONS

F78PB Solutions Semester 2 2023.

1.(a) $E(X^k) = M_X^{(k)}(0)$.

Put $k=2 \Rightarrow E(X^2) = M_X''(0)$ (1)

Now, $M_X = (1-10\theta)^{-2}$

$M_X' = -2(1-10\theta)^{-3} \cdot -10$

$= 20(1-10\theta)^{-3}$ (1)

And $M_X'' = -60(1-10\theta)^{-4} \cdot -10$

$= 600(1-10\theta)^{-4}$

Replace $\theta = 0$ into this to obtain

$M_X''(0) = 600 = E(X^2)$ (1)

Differentiate again.

$M_X''' = -2400(1-10\theta)^{-4} \cdot -10$

$= 24\,000(1-10\theta)^{-4}$

$M_X'''(0) = 24000 = E(X^3)$ (1)

(b) (i) $M_Y(\theta) = E(e^{\theta Y}) = E(e^{-\theta \log X})$ (1)

$= E((e^{\log X})^{-\theta})$

$= E(X^{-\theta})$ (1)

(ii) $E(X^{-\theta}) = \int_0^1 x^{-\theta} \cdot 1 \, dx$ (1)

$= \left[\frac{x^{-\theta+1}}{(1-\theta)}\right]_0^1 = \frac{1}{1-\theta}$ (1)

This is the mgf of Exp(1) so by uniqueness the distribution of Y is Exp(1). (1)

(ii) $E\left((\log(x))^2\right) = E(Y^2)$.

Now $E(Y^2) = M_Y''(0)$

$M_Y' = -(1-\theta)^{-2}.-1 = (1-\theta)^{-2}$

$M_Y'' = -2(1-\theta)^{-3}.-1$

$= 2(1-\theta)^{-3}$

$\therefore M_Y''(0) = 2 = E(Y^2)$. (1)

2. (a) We need the mean of this distribution.

$$E(X) = \int_0^\theta x.\frac{2x+\theta}{2\theta^2}\,dx \quad (1)$$

$$= \frac{1}{2\theta^2}\int_0^\theta (2x^2+\theta x)\,dx$$

$$= \frac{1}{2\theta^2}\left[\frac{2}{3}x^3 + \frac{1}{2}\theta x^2\right]_0^\theta$$

$$= \frac{1}{2\theta^2}\left\{\frac{2}{3}\theta^3 + \frac{1}{2}\theta^3\right\} - 0$$

$$= \frac{1}{2\theta^2}\cdot\frac{7\theta^3}{6} = \underline{\frac{7}{12}\theta} \quad (1)$$

(i) Bias $(\hat{\theta}) = E(\hat{\theta}) - \theta$ (1)

$= E(2\bar{X}) - \theta$

$= 2E(\bar{X}) - \theta$

$= 2E(X) - \theta$

$= \frac{14}{12}\theta - \theta$

$= \frac{2}{12}\theta = \underline{\frac{1}{6}\theta}$ (1)

(ii) For variance, we need $E(X^2)$

$E(X^2) = \int_0^\theta x^2 \cdot \frac{2x+\theta}{2\theta^2}\, dx$ (1)

$= \frac{1}{2\theta^2}\int_0^\theta (2x^3 + \theta x^2)\, dx$

$= \frac{1}{2\theta^2}\left[\frac{1}{2}x^4 + \frac{\theta}{3}x^3\right]_0^\theta$ (1)

$= \frac{1}{2\theta^2}\left\{\frac{1}{2}\theta^4 + \frac{1}{3}\theta^4\right\}_0^\theta$

$= \frac{1}{2\theta^2} \cdot \frac{5\theta^4}{6}$ (1)

$= \frac{5}{12}\theta^2$

So variance $= E(X^2) - [E(X)]^2$

$= \frac{5}{12}\theta^2 - \frac{49}{144}\theta^2$

$= \frac{11}{144}\theta^2$ (1)

$$MSE(\hat{\theta}) = var(\hat{\theta}) + bias^2(\hat{\theta}).$$

Now $var(\hat{\theta}) = var(2\bar{X}) = 4\, var(\bar{X}) = \frac{4}{n} var(X)$ (1)

This gives $MSE(\hat{\theta}) = \frac{44}{144n}\theta^2 + \left(\frac{1}{6}\theta\right)^2$

$$= \frac{44}{144n}\theta^2 + \frac{1}{36}\theta^2 \quad (1)$$

(b) Sample Mean is $\bar{X}$

Population Mean is $\frac{7}{12}\theta$ (1)

Equate the two for method of moments.

$$\bar{X} = \frac{7}{12}\hat{\theta}$$

$$\Rightarrow \hat{\theta} = \frac{12\bar{X}}{7}. \quad (1)$$

3. (a) Contribution to the likelihood function for those who have given birth

$$\beta e^{-\beta t_i}$$

Contribution for those who have still to give birth $\int_{270}^{\infty} \beta e^{-\beta t}\, dt$ (1)

$$= \left[-\frac{\beta}{\beta} e^{-\beta t} \right]_{270}^{\infty}$$

$$= 0 + e^{-270\beta}$$

$$= e^{-270\beta}$$ There are 2 women (1)

So $$L(\beta) = \prod_{i=1}^{10} \beta e^{-\beta t_i} \cdot \left(e^{-270\beta}\right)^2$$

$$= \beta^{10} \cdot e^{-\beta \Sigma t_i} \cdot e^{-270 \times 2\beta}$$ (1)

$$= \beta^{10} e^{-2673\beta} \cdot e^{-540\beta}$$

$$= \beta^{10} e^{-3213\beta}$$ (1)

(b) $$l(\beta) = \log\left\{\beta^{10} e^{-3213\beta}\right\}$$

$$= 10 \log\beta - 3213\beta$$ (1)

Differentiate and equate to 0.

$$l'(\beta) = \frac{10}{\beta} - 3213$$ (1)

$$\Rightarrow \quad \hat{\beta} = \frac{10}{3213} = 0.00311$$ (1)

Note that $l''(\beta) = -\frac{10}{\beta^2} < 0$ so the value is indeed a maximum. (1)

4(a)

(i) $\dfrac{\hat{p}-p}{\sqrt{\frac{p(1-p)}{n}}}$ is a pivotal quantity because

- It is a function of p and the sample. (1)
- It has a known distribution that does not depend on p – the normal distribution with standardised parameters $\mu=0$ and $\sigma^2=1$. (1)
- It is a monotonic function of p (1)

(ii) $\hat{p} = \dfrac{63}{100} = 0.63$ (1)

95% C.I. is $\left(\hat{p} - 1.96\sqrt{\dfrac{0.63\times 0.37}{100}},\ \hat{p} + 1.96\sqrt{\dfrac{0.63\times 0.37}{100}}\right)$ (1)

$= (0.63 - 0.095,\ 0.63 + 0.095)$

$= (0.535,\ 0.725)$. (1)

(b) The 95% C.I. for σ_1^2/σ_2^2 is given by

$$\left(\frac{s_1^2/s_2^2}{F_{24,12,0.025}},\ \frac{s_1^2/s_2^2}{F_{24,12,0.975}}\right)$$ (1)

$$= \left(\frac{10/12}{3.019},\ \frac{10/12}{(F_{12,24,0.025})^{-1}}\right)$$ (1)

$$= \left(\frac{10/12}{3.019},\ \frac{10/12}{2.541^{-1}}\right)$$

$= (0.276,\ 2.118)$ (1)

5. (a) $H_0 : \mu_1 - \mu_2 = 0$ (1)

$H_1 : \mu_1 - \mu_2 \neq 0$

test statistic $t = \dfrac{(\bar{x}_1 - \bar{x}_2) - (\mu_1 - \mu_2)}{s_p \sqrt{\frac{1}{n_1} + \frac{1}{n_2}}}$

And $s_p^2 = \dfrac{(n_1 - 1)s_1^2 + (n_2 - 1)s_2^2}{n_1 + n_2 - 2}$

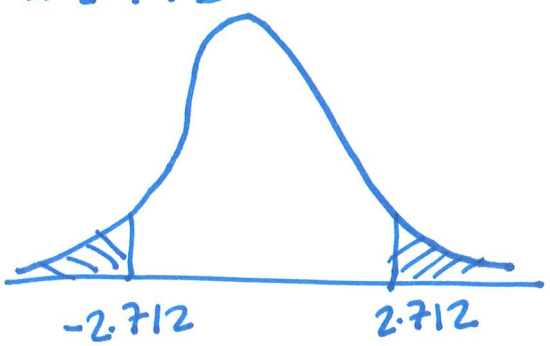
$= \dfrac{19 s_1^2 + 19 s_2^2}{38}$

$= \frac{1}{2}(s_1^2 + s_2^2) = \frac{1}{2}(61.73 + 62.94)$ (1)

$= 62.335.$

So $t = \dfrac{61.05 - 67.9}{\sqrt{62.335 \times \frac{1}{10}}} = \dfrac{61.05 - 67.9}{2.497}$

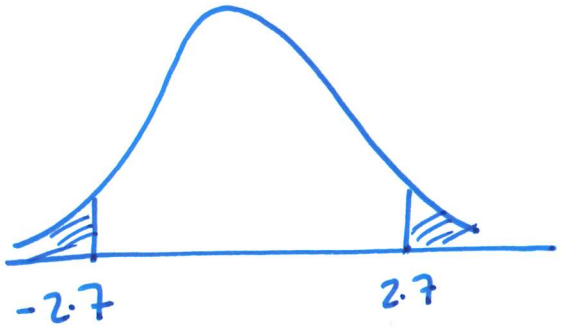
$= -2.743$ (1)



t_{38} curve. at 1%

t is in critical region (just) so we reject H_0. There is evidence of a difference between the two techniques. (1)

(b)



Using tables with $\nu = 40$ and $t = 2.7$ (1)
the approximate p-value is

$$2(1 - 0.9949) = 0.0051 \quad (1)$$

This is strong evidence against H_0 so, as in part (a), we have that there is a difference between the techniques. (1)

(c) We require data to be normally distributed (1) with equal population variances. (1).

6. (a) There is a positive association between yield of maize and amount of fertilizer used. (1).

(b) $S_{xy} = \sum xy - \frac{1}{n}\sum x \sum y$

$= 30284 - \frac{1}{20} 1119 \times 485$ (1)

$= 3148.25$

$S_{xx} = \sum x^2 - \frac{1}{n}(\sum x)^2$

$= 68807 - \frac{1}{20} \cdot (1119)^2$

$= 6198.95.$ (1)

$$S_{yy} = \Sigma y^2 - \frac{1}{n}(\Sigma y)^2$$
$$= 13799 - \frac{1}{20}(485)^2 \quad (1)$$
$$= 2037.75$$

$$r = \frac{S_{xy}}{\sqrt{S_{xx} S_{yy}}} = \frac{3148.25}{\sqrt{6198.95 \times 2037.75}}$$
$$= \underline{0.886} \quad (1)$$

(c) $H_0: \rho = 0$ (1)
$H_1: \rho \neq 0.$

test statistic $t = \dfrac{r\sqrt{n-2}}{\sqrt{1-r^2}}$ (1)

$= 8.107$

This follows a t_{18} distribution

critical region for $\alpha = 0.001$ is 3.922

or $t > 3.922$ Very strong evidence against H_0. (1)

(d) $\hat{\beta} = \dfrac{S_{xy}}{S_{xx}} = \dfrac{3148.25}{6198.95} = 0.508$ (1)

$\hat{\alpha} = \frac{1}{20}(485 - 0.508 \times 1119) = -4.173$

Equation $y = -4.173 + 0.508x$ (1)

(e) The graph looks fairly random (1) so the model seems to work reasonably well. (1).